
MODEL SELECTION AND VALIDATION REPORT (only_kde_fric_biomass model)

OBSERVATIONS:3382

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (only_kde_fric_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	7.08	0.17
kde_fric	-0.10	0.02
sbt_mean	0.31	0.06
sic_mean	0.05	0.05
sbt_sd	0.13	0.03
sst_sd	0.06	0.03
log_chla_mean	0.03	0.03
depth	-0.04	0.03

Dispersion parameter: 0.82

Spatiotemporal AR1 correlation (rho): 0.87

Matérn range: 264.77

Spatiotemporal marginal AR1 SD: 0.86

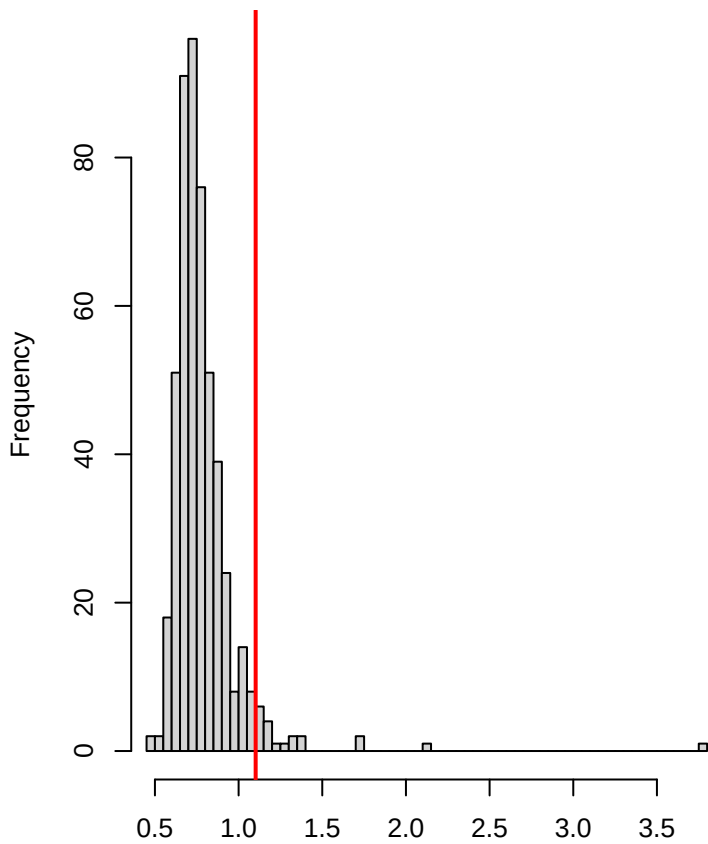
REML criterion at convergence: 27351.087

See ?tidy.sdmTMB to extract these values as a data frame.

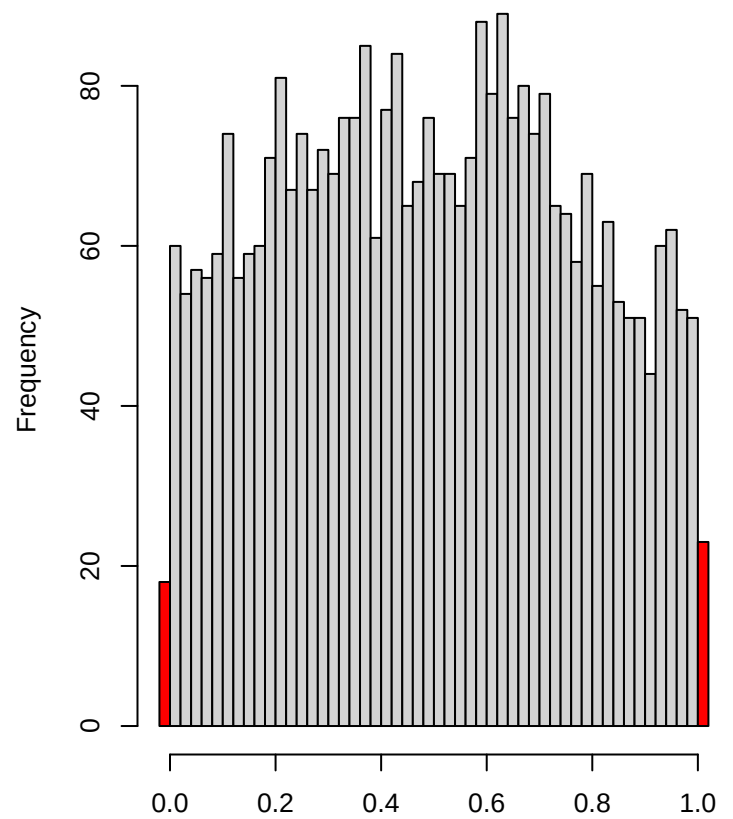
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	12	54726.17	3.504561
spatiotemporal_random_fields_ar1_with_anisotropy	14	54729.68	241.905178
spatiotemporal_random_fields_iid	11	54971.58	3.573743
spatiotemporal_random_fields_iid_with_anisotropy	13	54975.16	288.086723
spatial_random_fields	11	55263.24	3.537571
spatial_random_fields_with_anisotropy	13	55266.78	533.977433
error_only	9	55800.76	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



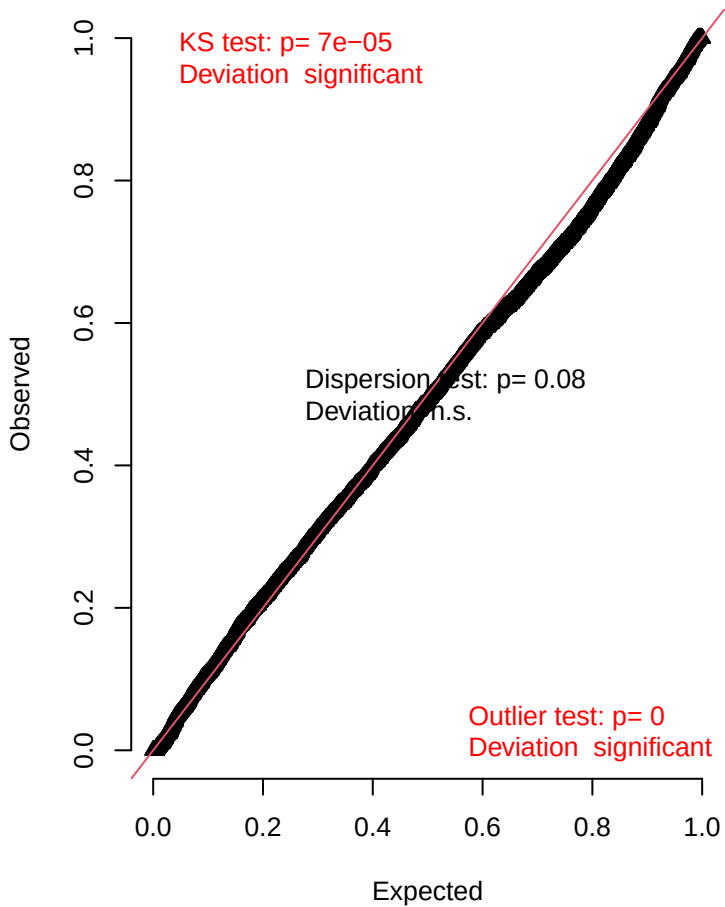
Outlier test significant



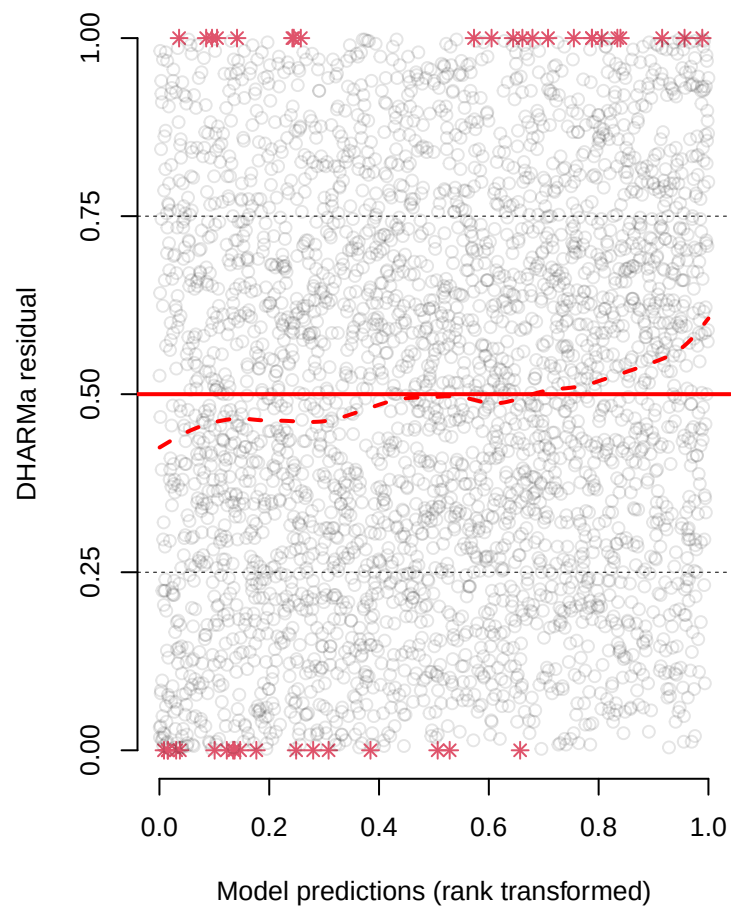
Simulated values, red line = fitted model. p-value (two.sided) :

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.014 | H1: Distance-based autocorrelation | p-value = 0.001

